EE 66 Fall 2026 — Signals, Dynamics, and Information — Homework 11

This homework is due Friday, November 13, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

On occasion, a problem set contains one more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

1. Simulating a Discrete-Time LTI System

You might think of a *system* as a function whose inputs and outputs are *both* signals in their entirety. It is common to express the relation between the inputs and outputs of an LTI system with the convolution sum,

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k],$$

but this is not the *only* way to do so. Any rule relating the input and output of the system works! For example, consider the "doubler" system, whose output is simply the input doubled. We can describe this system as

$$y[n] = 2x[n].$$

It turns out that this is a (quite simple, albeit) instance of a **linear constant coefficient difference equation** (LCCDE). An LCCDE is another common method used to express the relationship between the input and output of a system. In general, these will look something like

$$y[n] = \sum_{k=0}^{\infty} \alpha_k x[n-k] + \sum_{l=1}^{\infty} \beta_l y[n-l].$$

Notice the particular choice of starting index in each sum, chosen so that the output at time n depends only on the current input, past inputs, and past outputs. The coefficients α_k and β_l are of the same type as $y[n]$; most commonly, we deal with real valued signals, so $\alpha_k \in \mathbb{R}$ and $\beta_l \in \mathbb{R}$.

In this problem, we consider the discrete-time LTI system described by the LCCDE

$$y[n] = 0.8\,y[n-1] - 0.15\,y[n-2] + x[n] + 0.5\,x[n-1],$$

for all $n \in \mathbb{Z}$, with $y[n] = 0$ and $x[n] = 0$ for all $n < 0$, unless otherwise specified. In this problem you will use the attached notebook to explore several properties of this system; do *not* use any built-in "black-box" system functions (such as `lfilter` or built-in convolution) to compute the main answers unless explicitly instructed.

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(a) Write code to generate the impulse response $h[n]$ of the system for $n = 0, 1, \ldots, 50$ by choosing the input $x[n] = \delta[n]$. Plot $h[n]$ versus n, and based on the values and plot of $h[n]$, briefly explain whether this system behaves like an FIR or an IIR filter.

(b) Using the impulse response $h[n]$ from part (a), write a function that computes the output $y[n]$ for any finite-length input $x[n]$ using discrete-time convolution. Let the input be the finite-duration step

$$x[n] = \begin{cases} 1, & 0 \le n \le 100, \\ 0, & \text{otherwise}, \end{cases}$$

use your convolution function to compute the corresponding output $y_{\text{conv}}[n]$, then separately write code that computes $y[n]$ directly from the LCCDE to obtain $y_{\text{LCCDE}}[n]$, plot $y_{\text{conv}}[n]$ and $y_{\text{LCCDE}}[n]$ on the same graph, compute the maximum absolute difference between them, and briefly comment on why the two methods do or don't agree. Compute y_{conv} and y_{LCCDE} for $n = 0, 1, ..., 150$ and plot them.

(c) Using your finite-length approximation of $h[n]$ from part (a), write code to numerically approximate the frequency response $H(\omega) = \sum_{n=0}^{\infty} h[n]\, e^{-j\omega n}$ on a grid of frequencies $\omega \in [0, \pi]$. Plot $|H(\omega)|$ versus ω. Based on your plot, is the system more like a low-pass filter or a high-pass filter? Justify your answer.

(d) Let the input be the same as part (b). Using your LCCDE implementation, compute for $n = 0, \ldots, 100$:

(i) the total response $y_{\text{total}}[n]$ with initial conditions $y[-1] = 1$, $y[-2] = -0.5$;

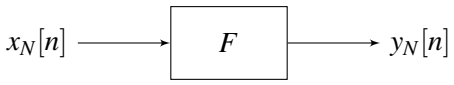
(ii) the zero-state response $y_{\text{ZSR}}[n]$ with initial conditions $y[-1] = y[-2] = 0$ and the same input $x[n]$;

(iii) the zero-input response $y_{\text{ZIR}}[n]$ with initial conditions $y[-1] = 1$, $y[-2] = -0.5$ and input $x[n] = 0$ for all n.

How do the zero-state and zero-input responses compare to the total response? Briefly comment on what happens to the zero-input response as n becomes large.

2. Discrete-Time Infinite Impulse Response (IIR) Filter (Adapted from EE 120 Homework)

Consider the causal, discrete-time LTI filter F shown below.



For each integer $N \geq 1$, define the input-output pair $(x_N[n], y_N[n])$ by the recursion

$$y_N[n] = \alpha y_N[n-N] + x_N[n],$$

where $|\alpha| < 1$. Assume the system starts at rest, i.e., for $n < 0$ we have $x_N[n] = 0$ and $y_N[n] = 0$.

(a) Let's first analyze the system for $N = 1$. For this part only, let $\alpha = 1$, and consider the step input $x_1[n] = u[n]$. Compute $y_1[0]$, $y_1[1]$, and $y_1[2]$ directly from the recursion. Then recognize the pattern and write a general expression for $y_1[n]$.

(b) Let $h_N[n]$ denote the impulse response of the N-system, i.e., the output when $x_N[n] = \delta[n]$. Determine an expression for $h_N[n]$ in terms of α and N. To do so, you should consider the case where n is or is not a multiple of N.

(c) Use your expression for $h_N[n]$ to sketch the impulse response in each of the following cases. Provide well-labeled sketches. Then briefly describe respectively the effect of changing α and N on the system's impulse response.

i. $\alpha = \frac{1}{2}$ and $N = 2$,
ii. $\alpha = \frac{1}{\sqrt{2}}$ and $N = 2$,
iii. $\alpha = \frac{1}{2}$ and $N = 4$.

(d) For the case $N = 3$, draw a block diagram with three separate first-order delays D, and label all internal signals. Then write the system in state-space form,

$$\mathbf{q}[n+1] = A\mathbf{q}[n] + Bx_3[n], \qquad y_3[n] = C\mathbf{q}[n] + D_u x_3[n],$$

and list the matrices A, B, C, and the scalar D_u.

Hint: Pick the output of every delay element as a state variable.

(e) Recall that a discrete-time LTI system is BIBO stable if and only if $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$. Use your expression for $h_N[n]$ from part (c) to compute $\sum_n |h_N[n]|$. For which values of α is the system BIBO stable?

3. Properties of Discrete-Time LTI Systems

(a) Consider a system described by the LCCDE

$$y[n] = \sum_{k=0}^{M} b_k x[n-k],$$

where the input satisfies $|x[n]| \leq B$ for all n, for some finite constant B. That is, the input is *bounded*. Prove that this system is BIBO stable. In particular, use the triangle inequality to prove that the output satisfies

$$|y[n]| \leq B \sum_{k=0}^{M} |b_k| \quad \text{for all } n.$$

Conclude that all systems determined by an FIR impulse response are BIBO stable. *Hint: determine the impulse response of the system described by this LCCDE and use the result above.*

(b) Consider a BIBO-stable LTI system with **real-valued** impulse response $h[n] \in \mathbb{R}$ for all n. Its frequency response is

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n] e^{-i\omega n}.$$

(i) Prove that the magnitude of the frequency response is finite for all ω; that is, show that $|H(\omega)| < \infty$ for every $\omega \in \mathbb{R}$. You may assume that the system is BIBO-stable.

(ii) Prove that the frequency response satisfies the conjugate symmetry property

$$H(-\omega) = H^*(\omega),$$

where * denotes complex conjugation.

(iii) Using this result, prove that $|H(\omega)|$ is an even function of ω and $\angle H(\omega)$ is an odd function of ω. That is, for all ω,

$$|H(\omega)| = |H(-\omega)| \quad \text{and} \quad \angle H(\omega) = -\angle H(-\omega).$$

(c) Consider any BIBO-stable LTI system with real impulse response $h[n]$ and frequency response

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n] e^{-i\omega n}.$$

(i) Prove that the DC gain ($\omega = 0$) satisfies

$$H(0) = \sum_{n=-\infty}^{\infty} h[n].$$

(ii) Prove that the gain at frequency $\omega = \pi$ satisfies

$$H(\pi) = \sum_{n=-\infty}^{\infty} (-1)^n h[n].$$

(iii) A low-pass filter (LPF) typically satisfies $|H(0)| \approx 1$ and $|H(\pi)| \approx 0$. Sketch a four-tap FIR impulse response which implements a low-pass filter. This means $h[n] \neq 0$ for $n = 0, 1, 2, 3$ only.

(iv) A high-pass filter (HPF) typically satisfies $|H(0)| \approx 0$ and $|H(\pi)| \approx 1$. Sketch a four-tap FIR impulse response which implements a high-pass filter.

(v) Consider the IIR impulse response $h[n] = \alpha^n u[n]$, where $|\alpha| < 1$. For which values of α is this best described as a LPF? For which values of α is it more similar to a HPF? Using this and the previous parts, conclude that a low-pass filter tends to look like a constant signal, while a high-pass filter tends to look like an alternating signal. In other words, a LPF contains mostly low frequencies, while a HPF contains mostly high frequencies.

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4. The Raccoons Are Plotting

Ryan is convinved that the local raccoons are plotting against him. To find out what they are saying, he records their conversations and tries to decipher their speech. Unfortunately, his microphone also picks up a lot of background noise! Before he can uncover the raccoons' plans he wants to clean up his recordings.

Let $x[n]$ be Ryan's recording.

(a) Ryan remembers the two-point moving average filter he used to count raccoons, so he applies it to his recording. Its impulse response is

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1].$$

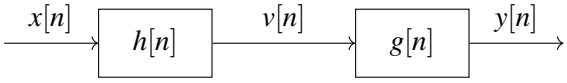
Find $H(\omega)$. Is this a high or low pass filter?

(b) After listening to the filtered recording, Ryan decides that some of the raccoons' suspicious squeaks must be hiding in the higher-frequency components that remain. He sends the moving average's output through a first-difference filter with impulse response

$$g[n] = \delta[n] - \delta[n-1].$$

Find its frequency response $G(\omega)$. What kind of filter is this?

(c) Ryan's two filters are now connected in **series**: the output of the first is the input to the second.

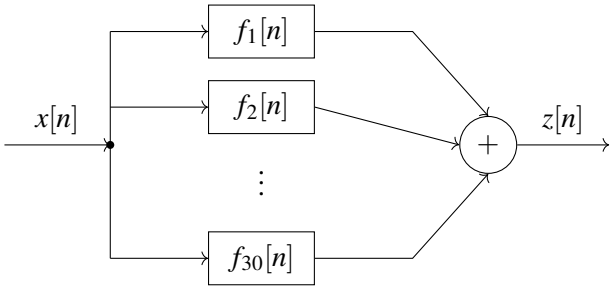


Find the impulse response $c[n]$ of the overall system. Then, find the overall frequency response $C(\omega)$ **using the new impulse response**. At which frequencies is the magnitude zero, and where is it largest? What kind of filter has Ryan actually built?

(d) Ryan wonders if he could have saved himself some work. Compute $H(\omega)G(\omega)$ using your answers from parts (a) and (b), and compare it with $C(\omega)$. What does this suggest about the frequency response of two LTI systems connected in series?

(e) Ryan wants to know if this property holds in general. Using the definition of the frequency response, prove that convolution of two impulse responses corresponds to multiplication of their frequency responses.

(f) For reasons yet unknown to Ryan, there are a lot of raccoons in town this week. He decides he needs to apply more powerful processing to each recording. He consults with speech experts at Berkeley and then builds a perfect LTI processing system with 30 filters working in **parallel**, followed by an adder. Every branch receives the same recording $x[n]$, and Ryan adds all 30 branch outputs. Let $f_i[n]$ be the impulse response of branch i, and let $F_i(\omega)$ be its frequency response, for $i = 1, \ldots, 30$.



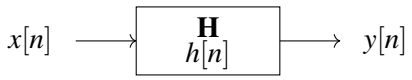
Find the impulse response $p[n]$ and frequency response $P(\omega)$ of this overall parallel system in terms of the individual branches. How does this differ from the series connection in part (c)?

5. Beep Boop

Nathan uses his radio transmitter to send a signal $x[n]$ to Aditya's receiver. Aditya hears $y[n]$. Ideally, Aditya hears exactly what Nathan sent. This is unfortunately not the case.

Nathan's signal travels in a straight line and arrives instantly. However, a second weaker copy bounces off of The Campanile and arrives two samples later at half amplitude. A third copy bounces off of a raccoon den in Strawberry Creek, which arrives three samples later at half amplitude. The receiver adds together everything that reaches its antenna, and Aditya hears a distorted signal.

We can model this as an LTI system:



(a) Using the description above, write $y[n]$ in terms of $x[n]$. Then determine the impulse response $h[n]$ of the communication channel and sketch it.

(b) Nathan then sends the signal

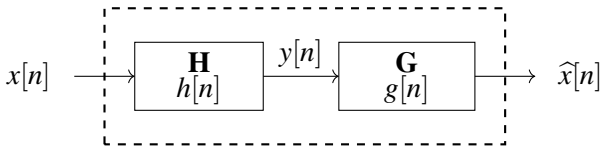
$$x[n] = 2\delta[n] + \delta[n-1] - \delta[n-2].$$

Does Aditya receive Nathan's signal clearly? Explain why or why not.

(c) To clean up the measurement, Aditya tries subtracting off the reflections. In particular, he passes $y[n]$ through a filter with impulse response

$$g[n] = \delta[n] - \tfrac{1}{2}\,\delta[n-2] - \tfrac{1}{2}\,\delta[n-3].$$

We can model the overall system as follows, with $\widehat{x}[n]$ the recovered signal heard by Aditya.



$\tilde{\mathbf{H}}$, overall system

Consider the *overall system* $\tilde{\mathbf{H}}$, with impulse response $\tilde{h}[n]$ such that $\widehat{x}[n] = (\tilde{h} * x)[n]$. Find $\tilde{h}[n]$ and apply it to the signal from part (b). Explain why Aditya doesn't recover Nathan's signal exactly.

(d) Aditya is now in search of a filter that will actually help receive Nathan's signal clearly, so he ventures into the frequency domain. Recall that we can use the eigenfunction property by feeding $x[n] = e^{i\omega n}$ into the channel to find its frequency response $H(\omega)$.

Find the frequency response $F(\omega)$ of the filter that undoes the distortion from the channel.

Hint: Suppose $x[n] = e^{i\omega n}$. What does this frequency look like after going through $\mathbf{H}$? What about at the output, after going through $\mathbf{F}$?

(e) Prove the *delay property* of DT LTI systems. That is, if $h[n]$ has frequency response $H(\omega)$, then $h[n-n_0]$ has frequency response $e^{-i\omega n_0}H(\omega)$.

(f) Use the previous parts to determine an LCCDE describing Aditya's recovery strategy. In this case, $y[n]$ is the input (received, corrupted signal) and $\widehat{x}[n]$ is the output (cleaned signal).

Big Hint: Use the frequency response to write something like $\alpha F(\omega) + \beta F(\omega) + \gamma F(\omega) = 1$. What happens to $y[n]$ when you multiply its constituent frequencies by each side of this equality? To get you started, multiplying $y[n]$'s frequencies by $F(\omega)$ is the same as convolving $y[n]$ with the system's impulse response, and multiplying by 1 is the same as convolving with $\delta[n]$.

6. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

(a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

(b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

(c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

(d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

(e) Is there anything else you would like to share?